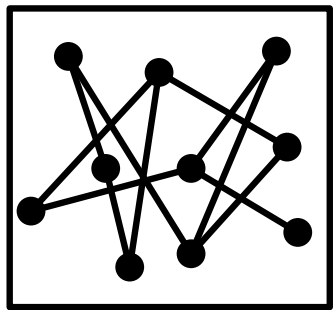
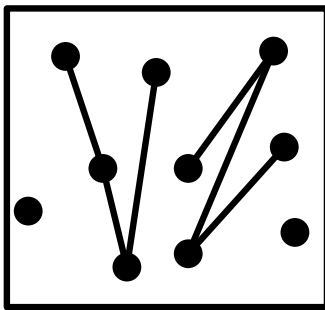


Planar Stories of Graph Drawings: Algorithms and Experiments

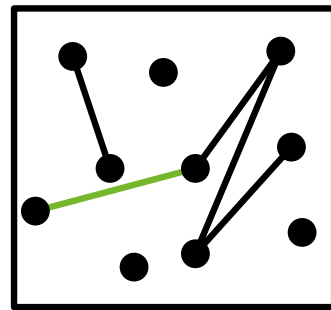
IN:



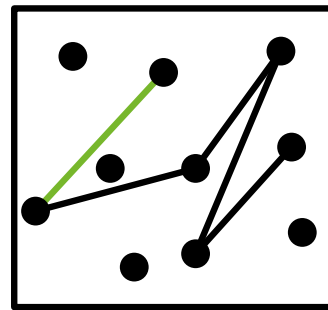
OUT:



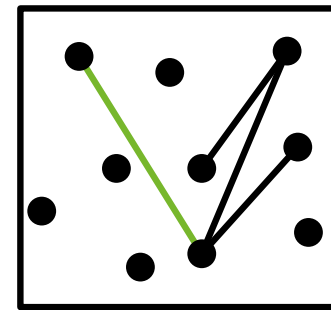
F_1



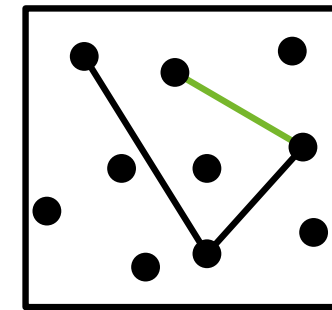
F_2



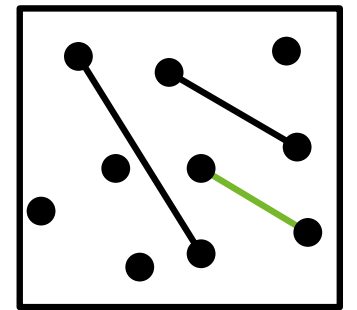
F_3



F_4



F_5



F_6

NP-hardness · Heuristics · ILP · Experiments

Carla Binucci¹, Sabine Cornelsen², Walter Didimo¹, Seok-Hee Hong³, Eleni Katsanou⁴, Maurizio Patrignani⁵, Antonios Symvonis⁴, **Samuel Wolf**⁶

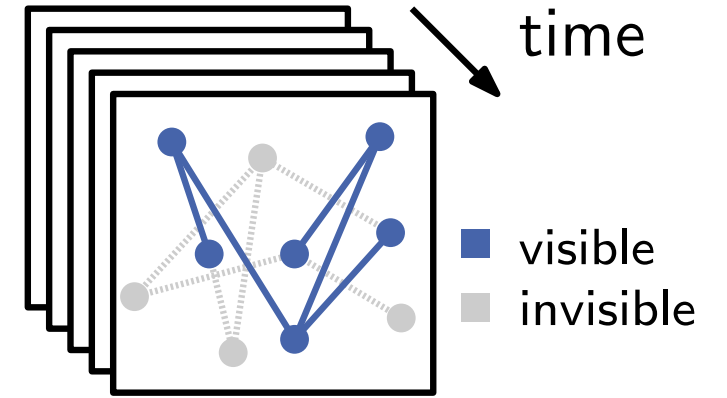
Project originated in Bertinoro Workshop '25

Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .



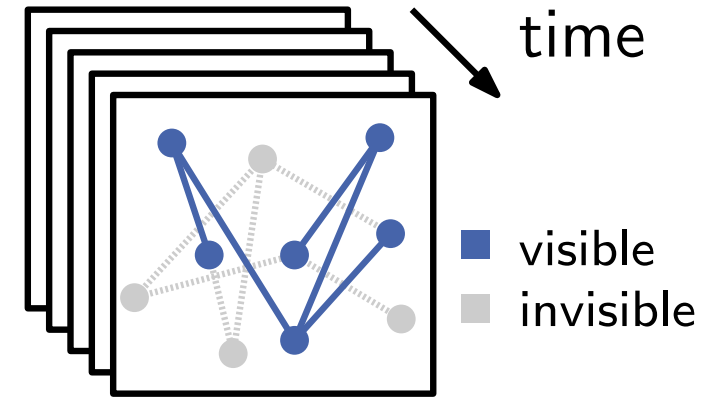
Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .

Often: Every vertex has the same position in all frames it appears in; (dis-)appears once

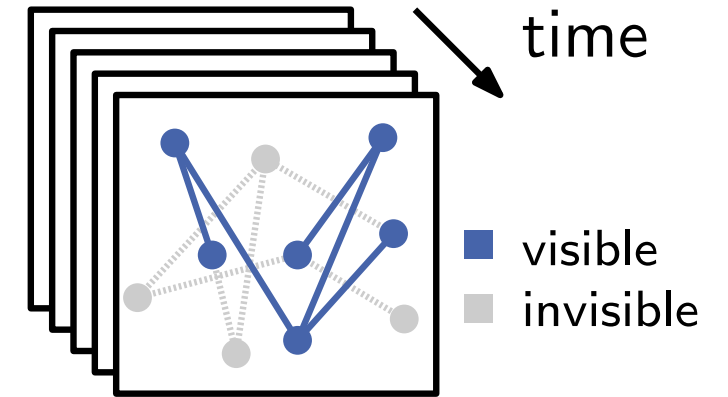


Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .

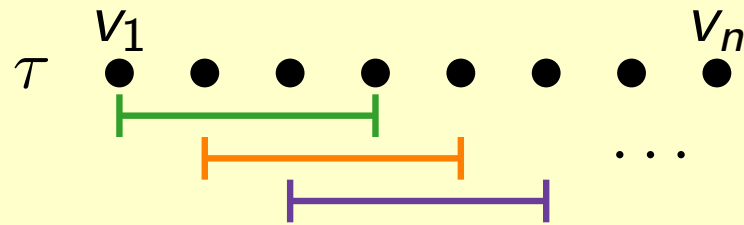


Often: Every vertex has the same position in all frames it appears in; (dis-)appears once

Graph Stories

[Borrizzo, Da Lozzo, Di Battista, Frati, Patrignani: JGAA'20]

Given: Graph G , vertex order τ , $W \in \mathbb{N}$.



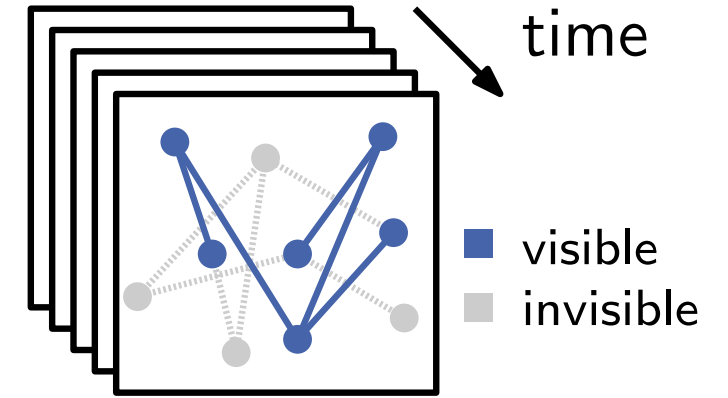
Want: planar straight-line story respecting τ with small area w.r.t. W in each frame.

Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .

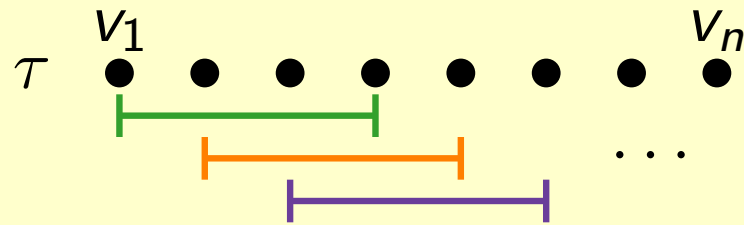


Often: Every vertex has the same position in all frames it appears in; (dis-)appears once

Graph Stories

[Borrizzo, Da Lozzo, Di Battista, Frati, Patrignani: JGAA'20]

Given: Graph G , vertex order τ , $W \in \mathbb{N}$.



Want: planar straight-line story respecting τ with small area w.r.t. W in each frame.

Story Plans

[Binucci, Di Giacomo, Lenhart, Liotta, Montecchiani, Nöllenburg, Symvonis: JCSS'24]

Given: Graph G .

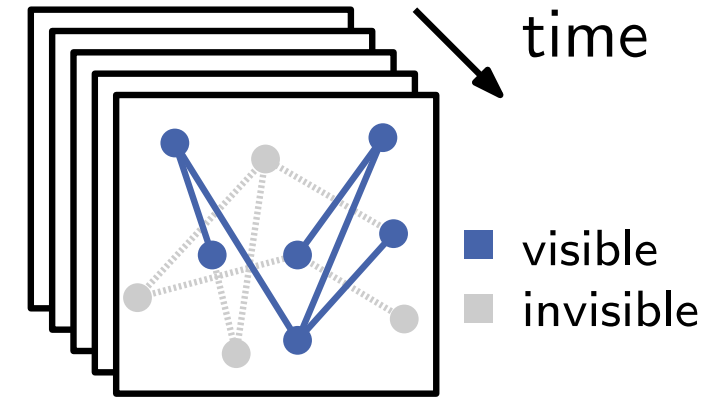
Want: planar straight-line story s.t. every vertex persists until all of its neighbours appeared.

Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .

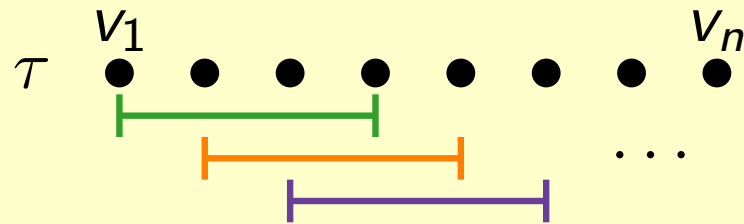


Often: Every vertex has the same position in all frames it appears in; (dis-)appears once

Graph Stories

[Borrizzo, Da Lozzo, Di Battista, Frati, Patrignani: JGAA'20]

Given: Graph G , vertex order τ , $W \in \mathbb{N}$.



Want: planar straight-line story respecting τ with small area w.r.t. W in each frame.

Story Plans

[Binucci, Di Giacomo, Lenhart, Liotta, Montecchiani, Nöllenburg, Symvonis: JCSS'24]

Given: Graph G .

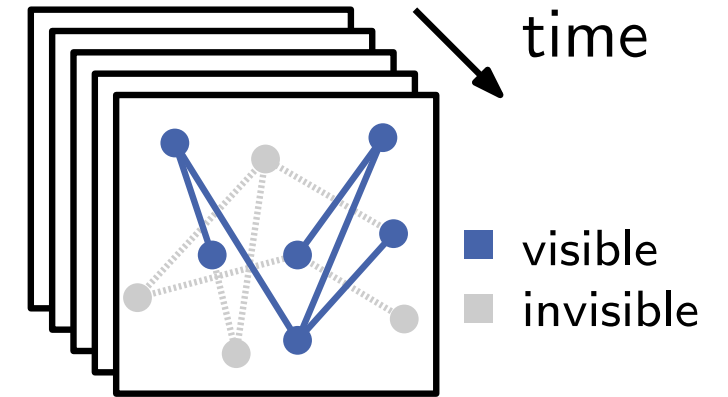
Want: planar straight-line story s.t. every vertex persists until all of its neighbours appeared.

Dynamic Visualization: Graph Stories and Story Plans

General Setting:

Given: A graph G .

Want: A *story*, i.e., a sequence of partial drawings (*frames*) of G such that the union of frames represents G .

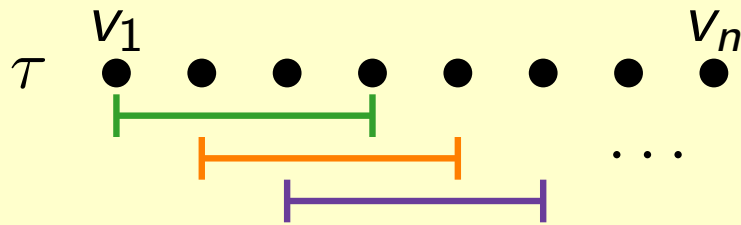


Often: Every vertex has the same position in all frames it appears in; (dis-)appears once

Graph Stories

[Borrizzo, Da Lozzo, Di Battista, Frati, Patrignani: JGAA'20]

Given: Graph G , vertex order τ , $W \in \mathbb{N}$.



Want: planar straight-line story respecting τ with small area w.r.t. W in each frame.

Story Plans

[Binucci, Di Giacomo, Lenhart, Liotta, Montecchiani, Nöllenburg, Symvonis: JCSS'24]

Given: Graph G .

Want: planar straight-line story s.t. every vertex persists until all of its neighbours appeared.

Outerplanar, forest, Jordan curves: [Fiala, Firman, Liotta, Wolff, Zink: SOFSEM'24]

Jordan curves, mapping on points: [Di Battista, Didimo, Grilli, Grosso, Ortali, Patrignani, Tappini: JGAA'23]

Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

Our Setting

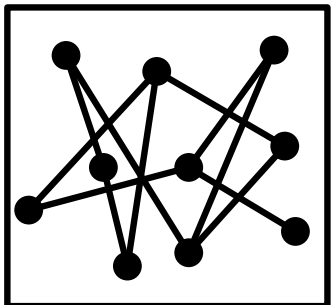
Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

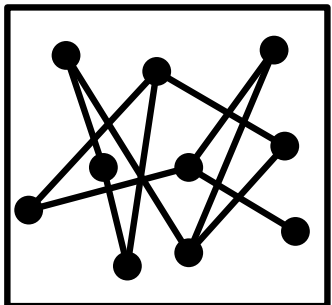
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

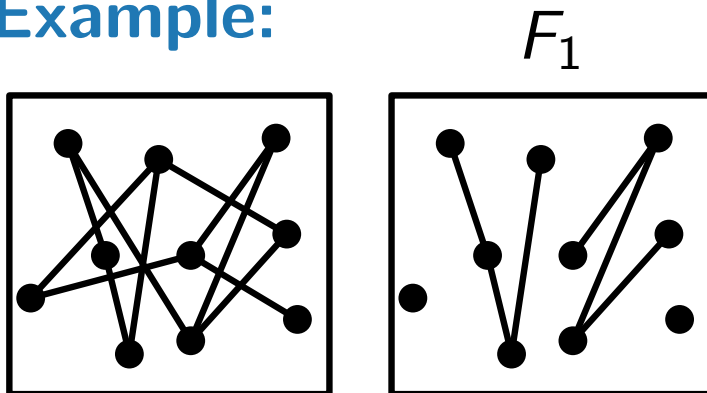
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

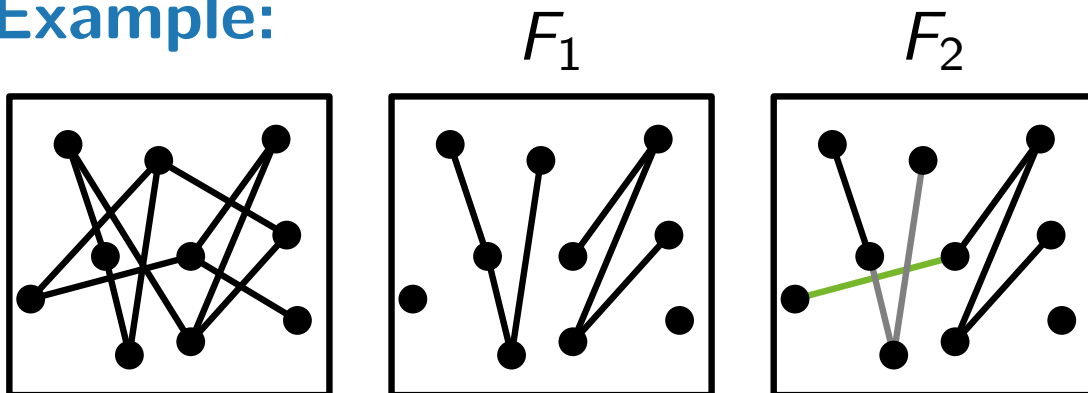
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

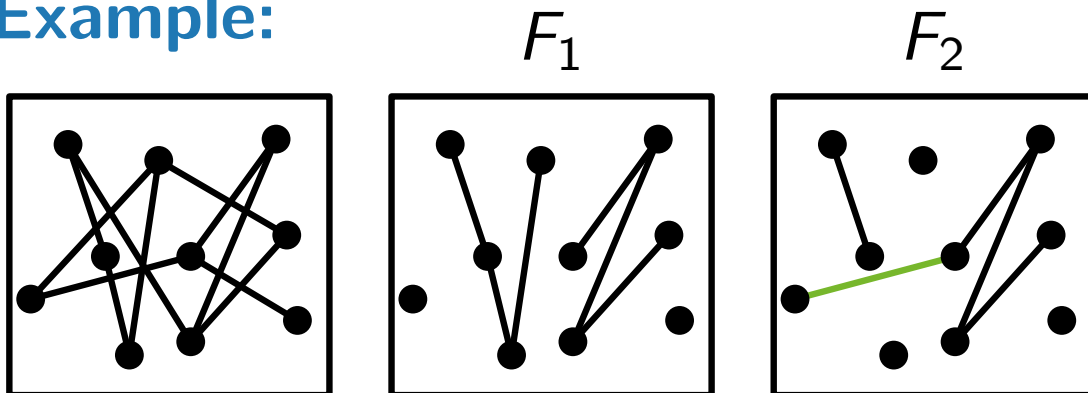
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

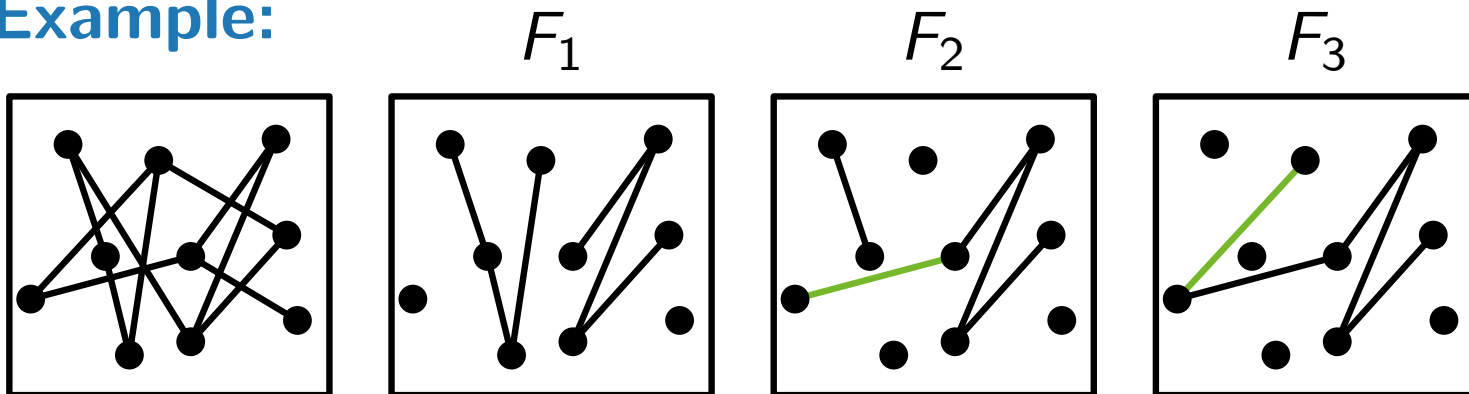
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

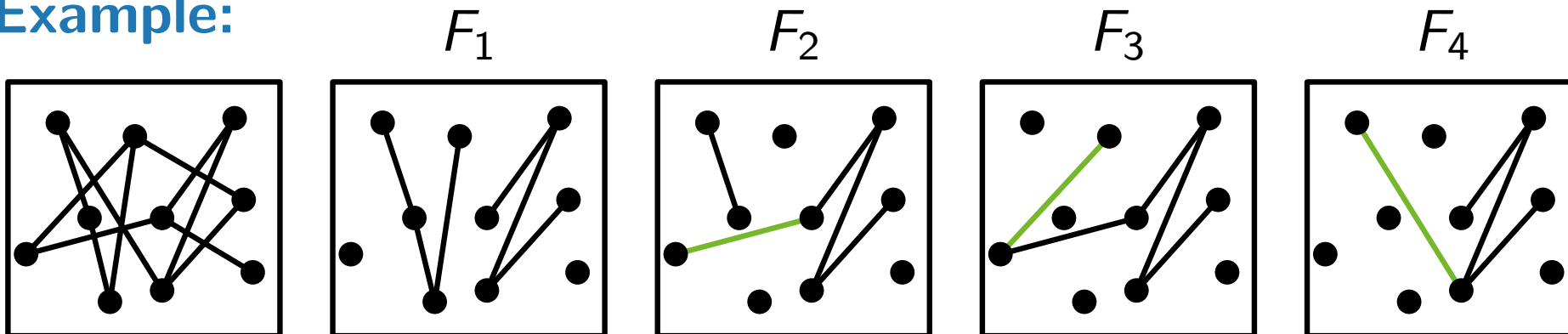
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

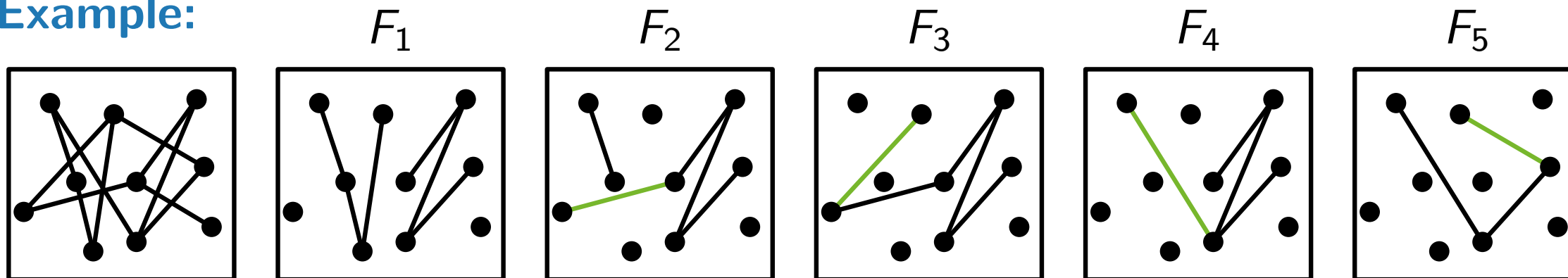
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

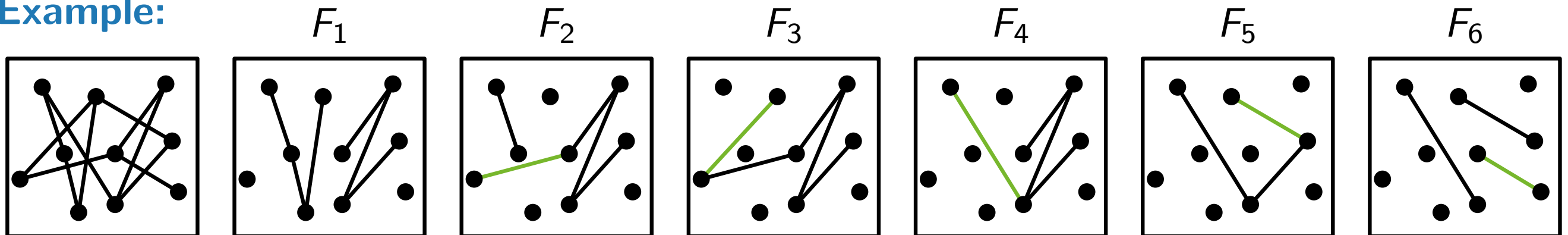
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

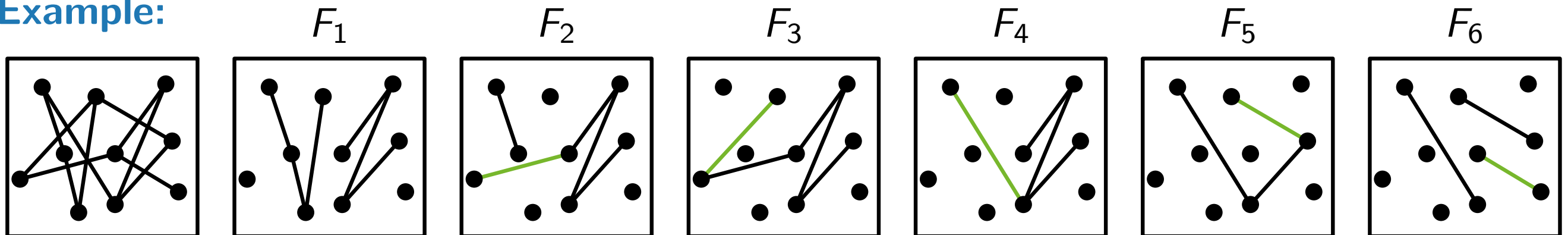
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.
- Once an edge disappears it cannot reappear.
- The minimum number of edges in a frame is maximized.

Example:



Our Setting

Idea: Use a similar approach to visualize complex *geometric* graphs with (many) crossings.

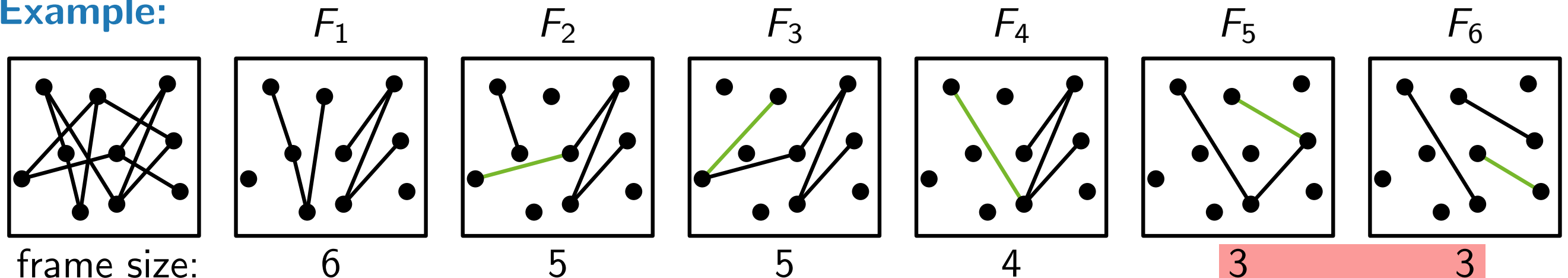
MAXMINFRAMEPLANARSTORY

Given: A geometric graph G .

Want: Planar (straight-line) story s.t.

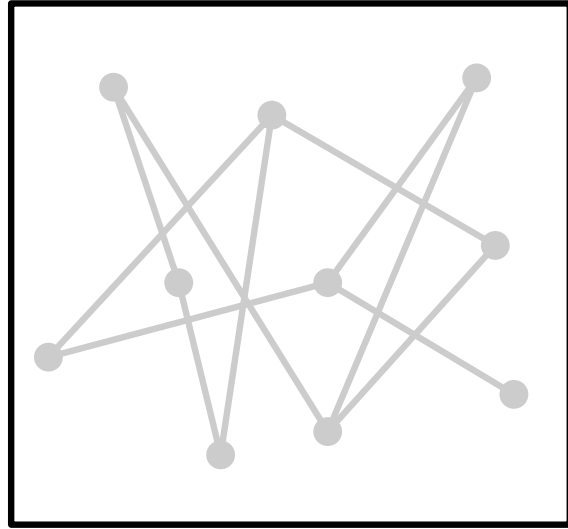
- All vertices of G are present in each frame at their initial positions.
- The initial frame contains a planar subgraph of G .
- In each subsequent frame exactly one new edge appears.
- Once an edge disappears it cannot reappear.
- The **minimum** number of edges in a frame is maximized.

Example:



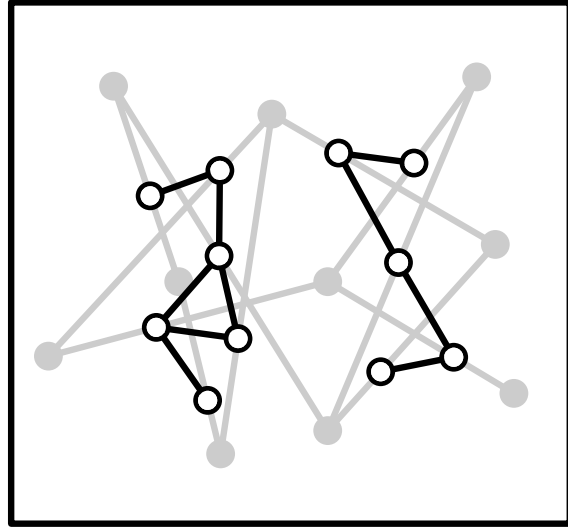
Our Setting: Crossing Graphs and Independent Sets

Crossing Graph:



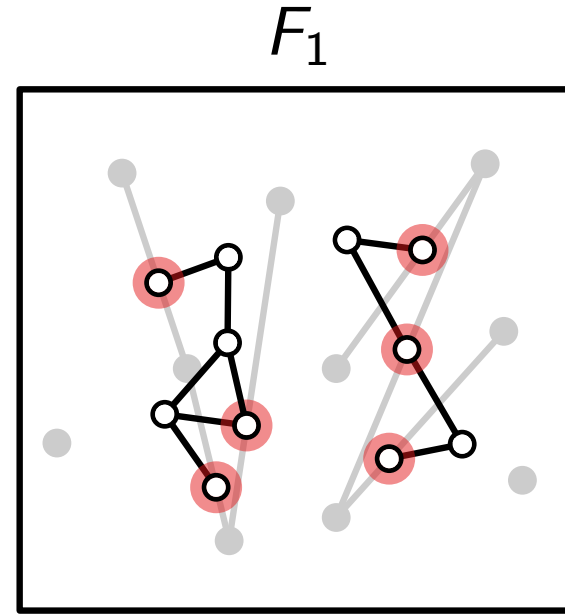
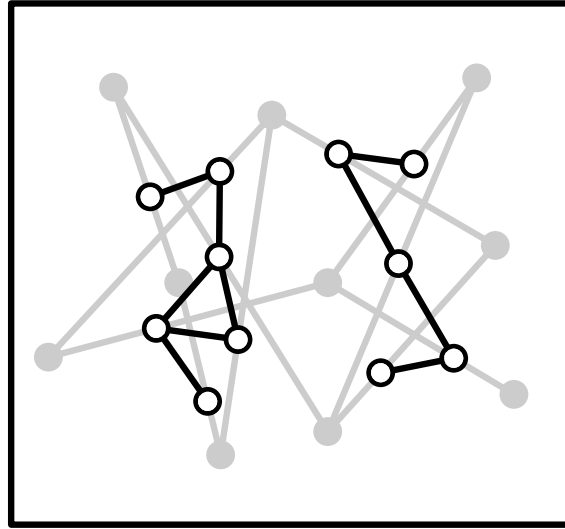
Our Setting: Crossing Graphs and Independent Sets

Crossing Graph:



Our Setting: Crossing Graphs and Independent Sets

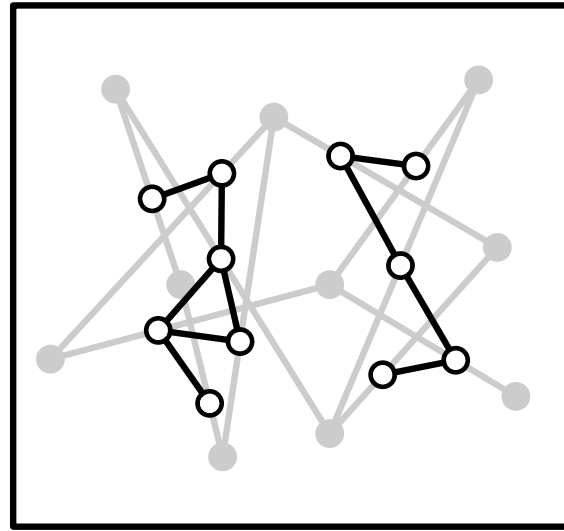
Crossing Graph:



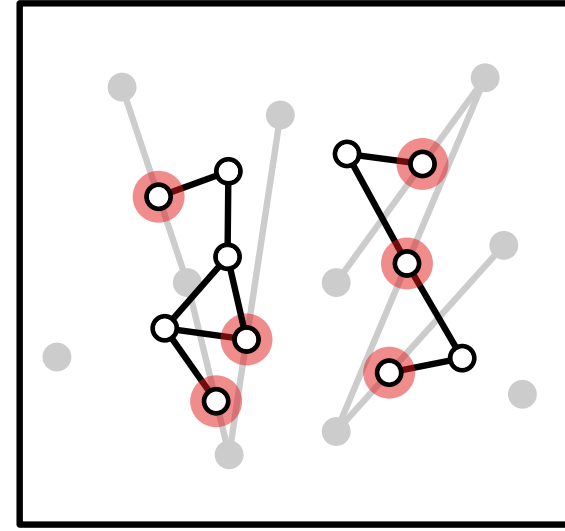
Observation: A planar frame corresponds to an independent set in the crossing graph.

Our Setting: Crossing Graphs and Independent Sets

Crossing Graph:



F_1



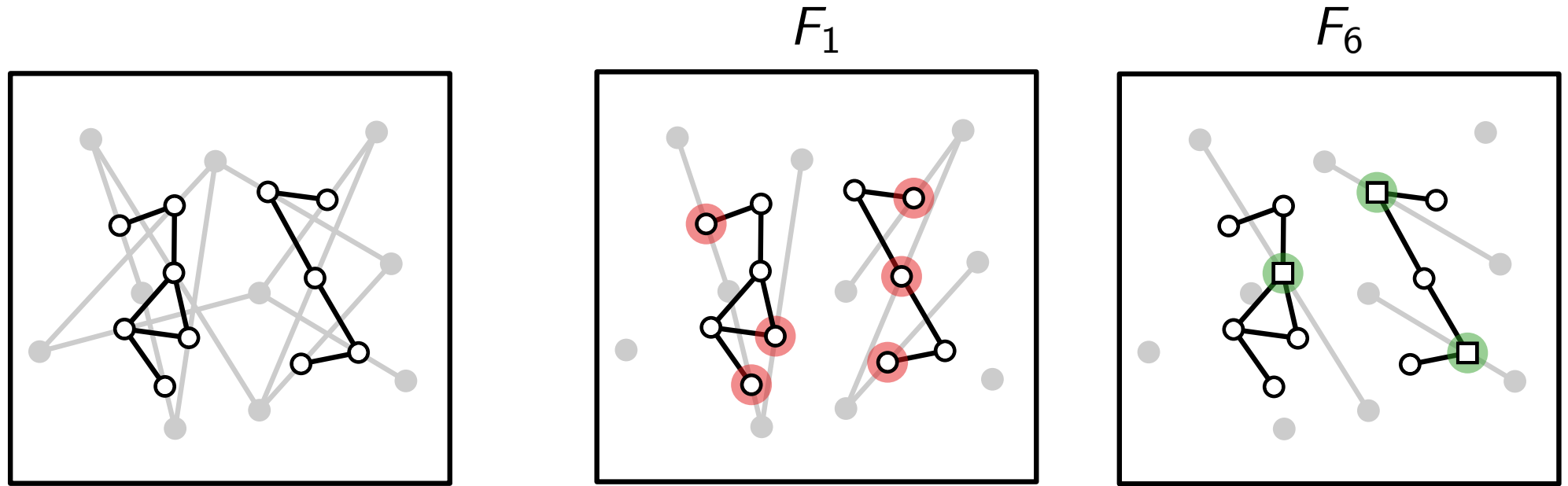
Observation: A planar frame corresponds to an independent set in the crossing graph.

Observation: Isolated vertices in the crossing graph can be present in every frame.

$\Rightarrow$ Assume the crossing graph has no isolated vertices.

Our Setting: Crossing Graphs and Independent Sets

Crossing Graph:



Observation: A planar frame corresponds to an independent set in the crossing graph.

Observation: Isolated vertices in the crossing graph can be present in every frame.

⇒ Assume the crossing graph has no isolated vertices.

⇒ Initial and final frame are disjoint.

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

Idea:

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

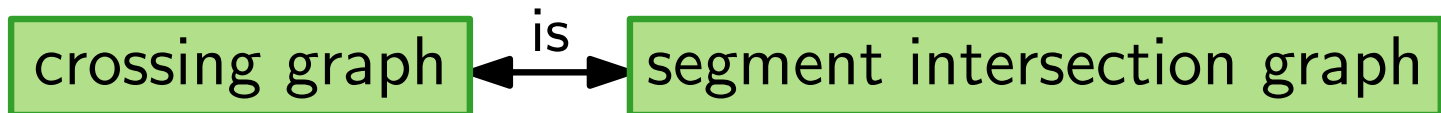
Idea: Not every graph is a crossing graph of a geometric graph.

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

Idea: Not every graph is a crossing graph of a geometric graph.



Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

⇒ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

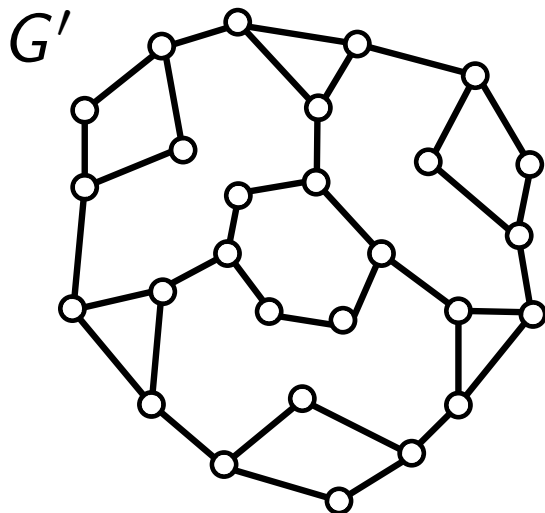
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

$\Rightarrow$ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

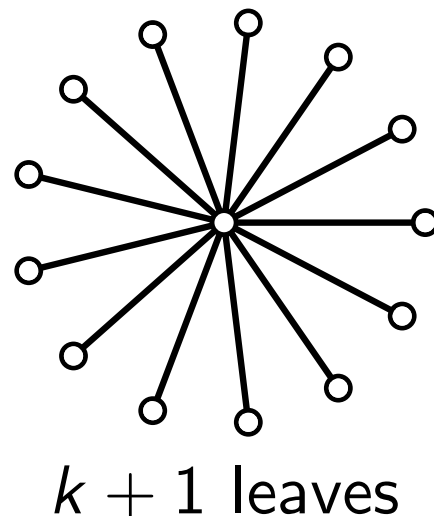
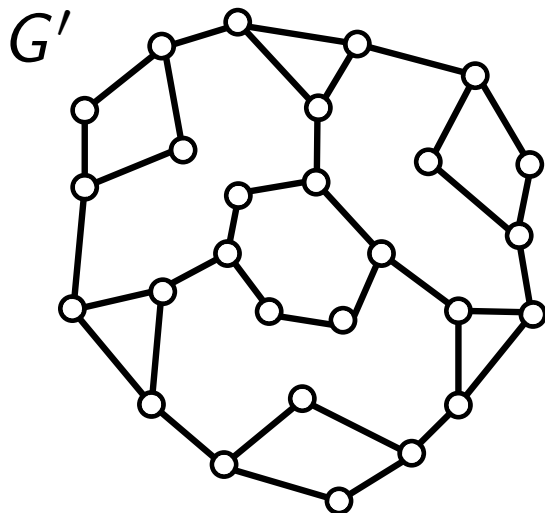
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

⇒ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

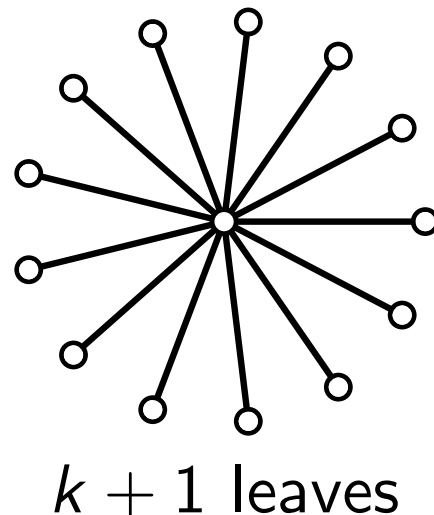
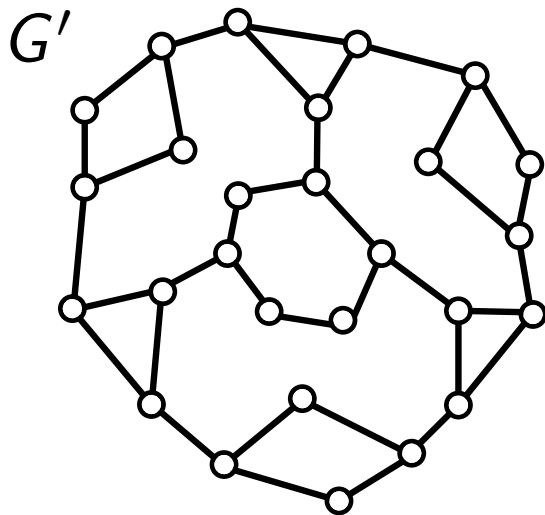
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

$\Rightarrow$ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



minimum frame size $k + 1 \Leftrightarrow$
independent set in G' of size k

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

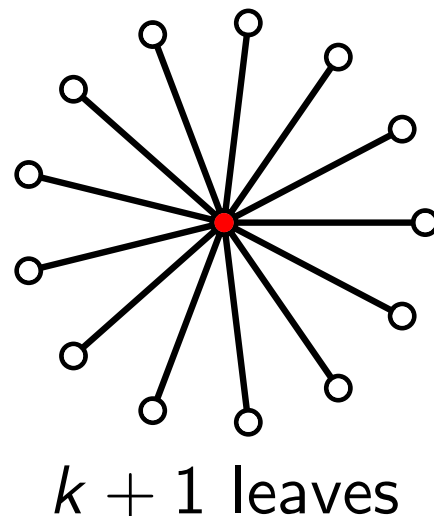
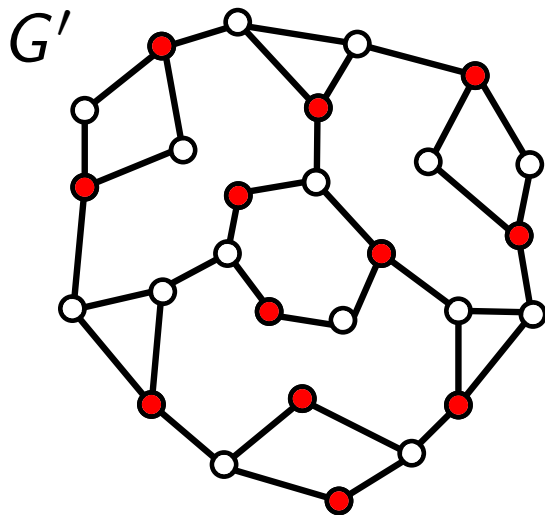
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

$\Rightarrow$ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



minimum frame size $k+1 \Leftrightarrow$
independent set in G' of size k

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

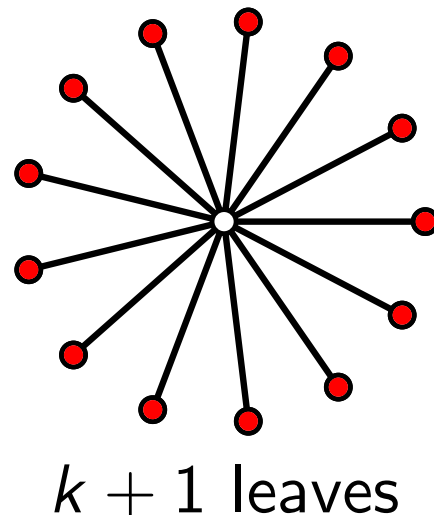
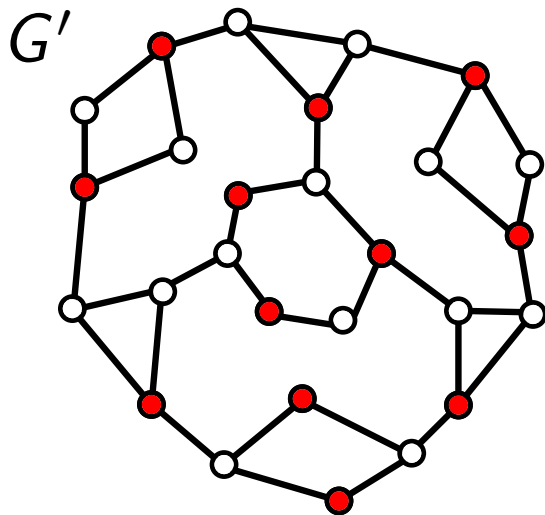
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

$\Rightarrow$ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



minimum frame size $k + 1 \Leftrightarrow$
independent set in G' of size k

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

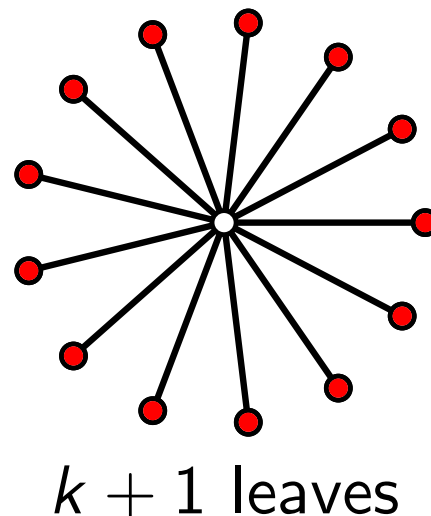
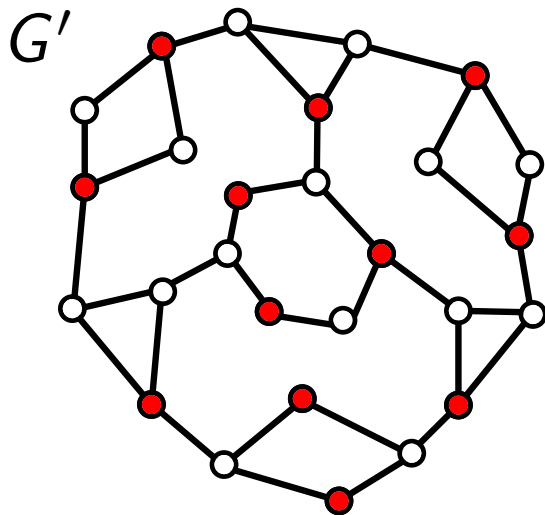
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

⇒ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



G is 2-planar
solvable in linear time

Problem Complexity

Theorem. MAXMINFRAMEPLANARSTORY is NP-hard.

That is: Given a geometric graph G is, it is NP-hard to decide whether there is a story whose minimum number of edges in a frame is at least ℓ .

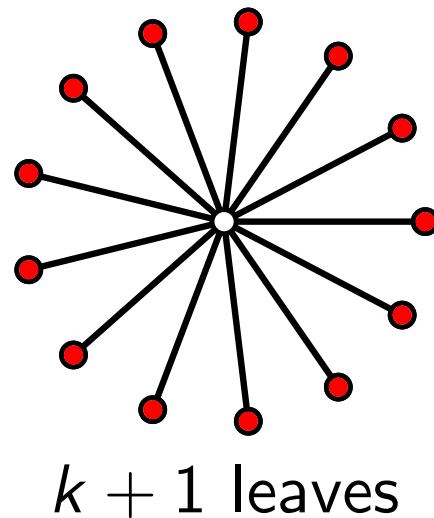
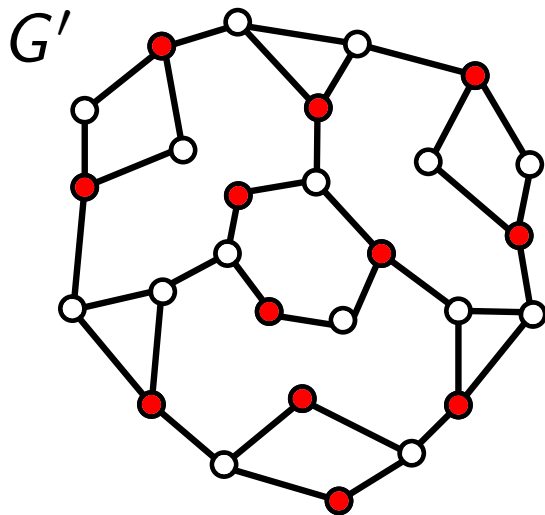
Idea: Not every graph is a crossing graph of a geometric graph.



[Chalopin, Gonçalves: STOC'09]

⇒ Reduce from PLANAR INDEPENDENT SET [Mohar: JCTB'01]

Given a planar graph G' , is there an independent set of size at least k ?



G is 2-planar
solvable in linear time

G is 3-planar
Crossing graph has no
cycles
solvable in cubic time

Greedy Heuristic

Simple Greedy

Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

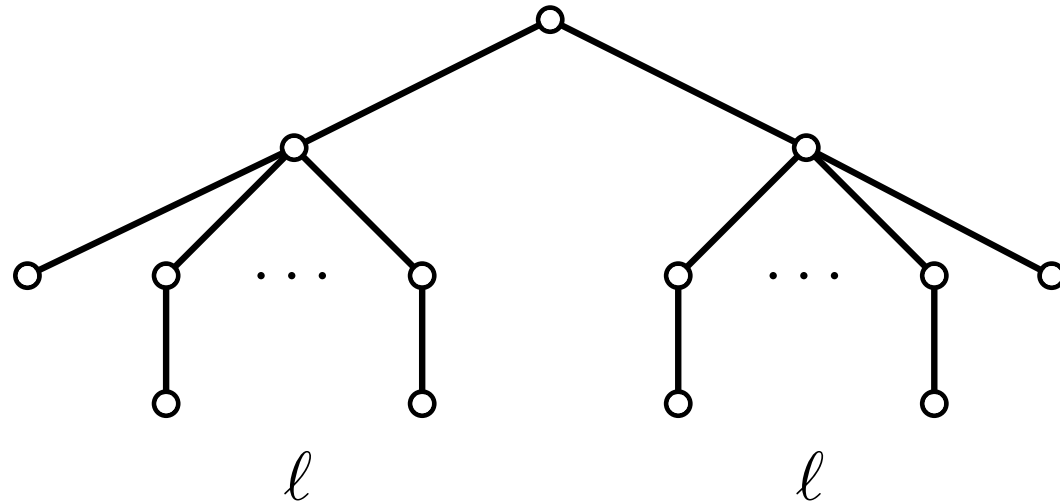
Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



Greedy Heuristic

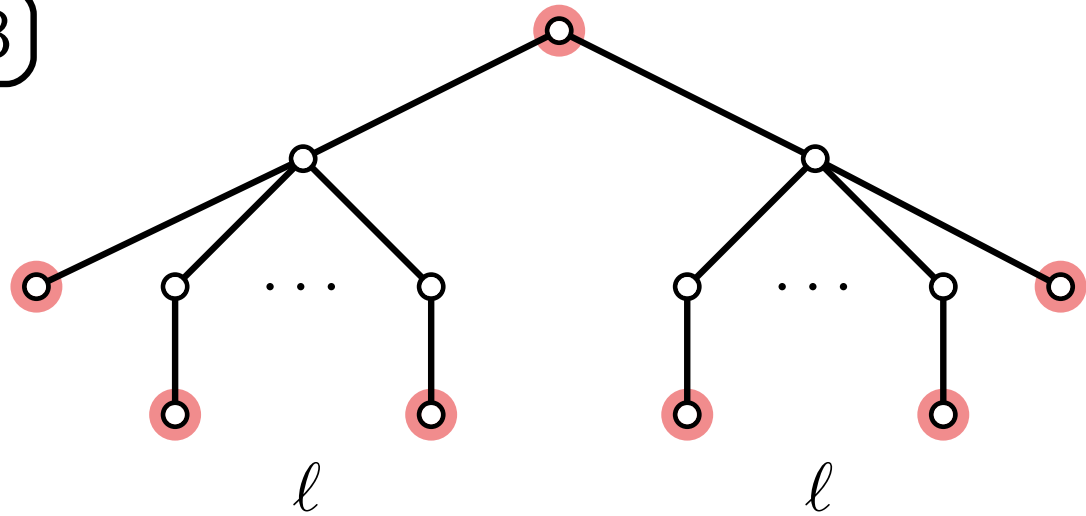
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

$$2\ell + 3$$



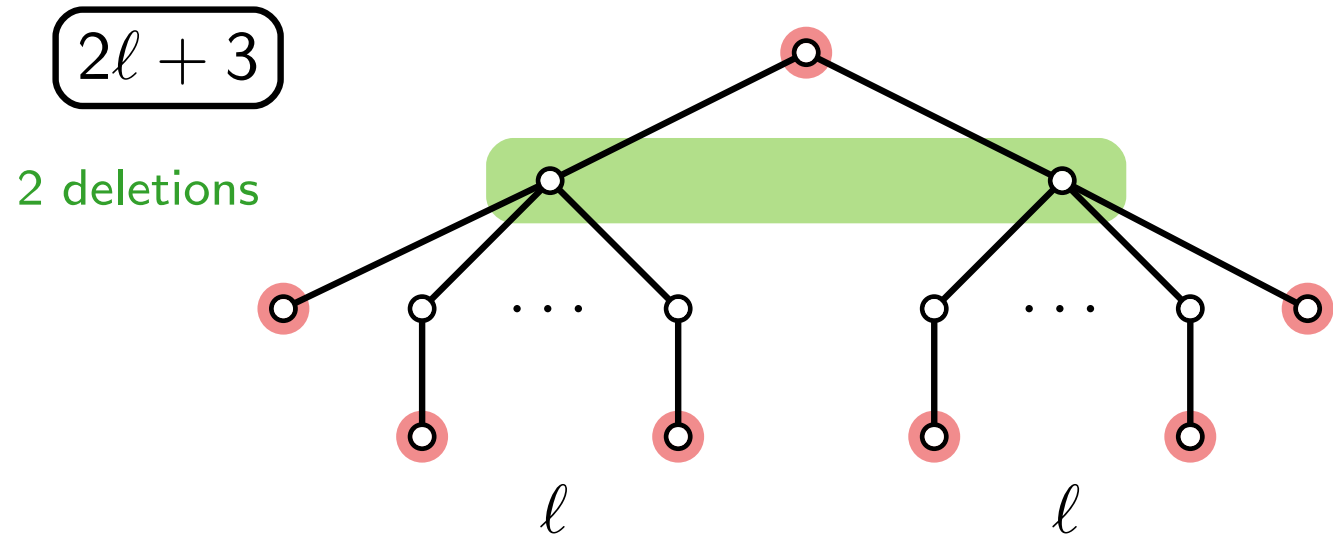
Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



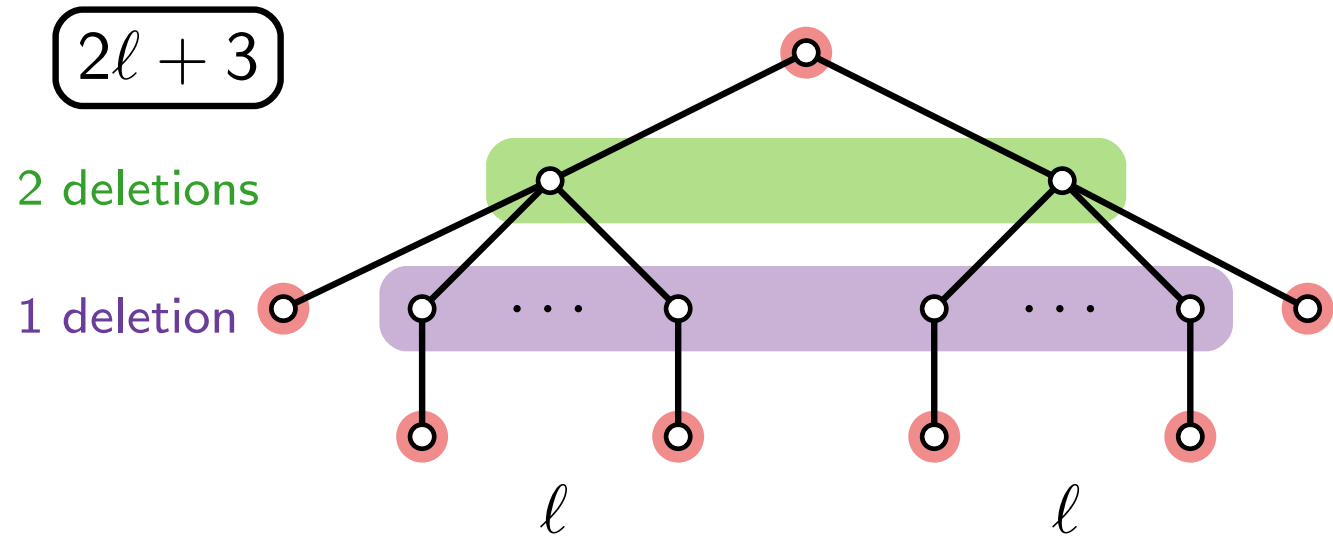
Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



Greedy Heuristic

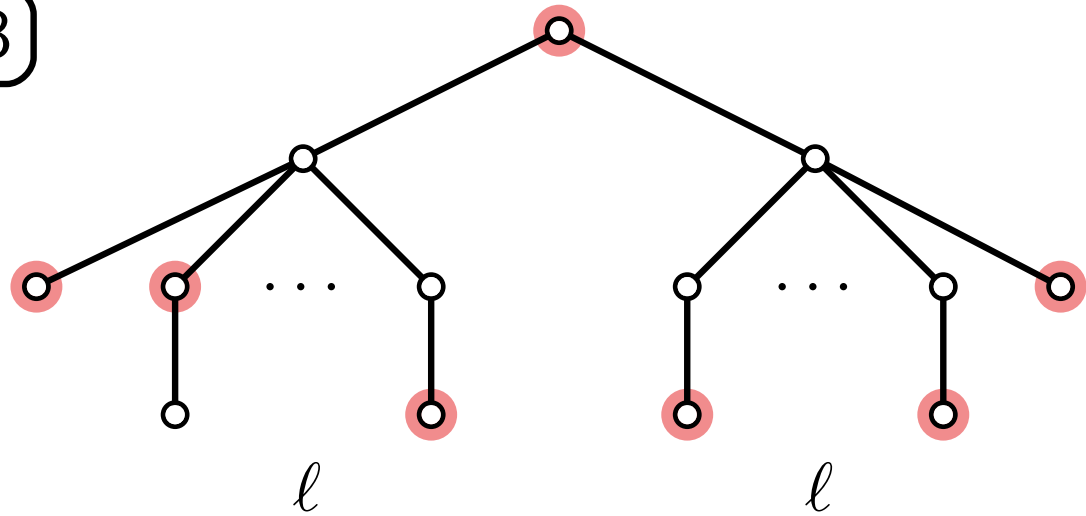
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

$$2\ell + 3$$



Greedy Heuristic

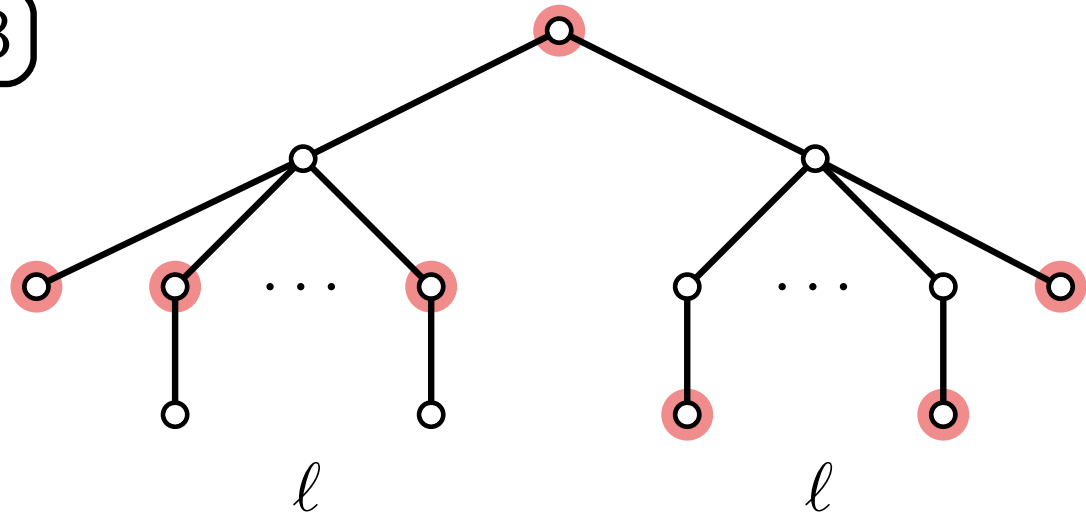
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

$$2\ell + 3$$



Greedy Heuristic

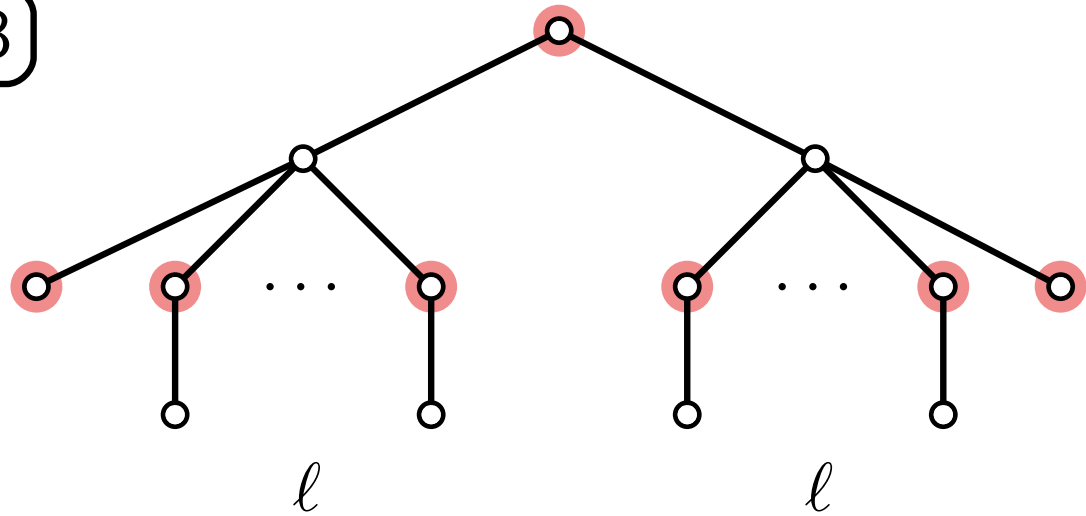
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

$$2\ell + 3$$



Greedy Heuristic

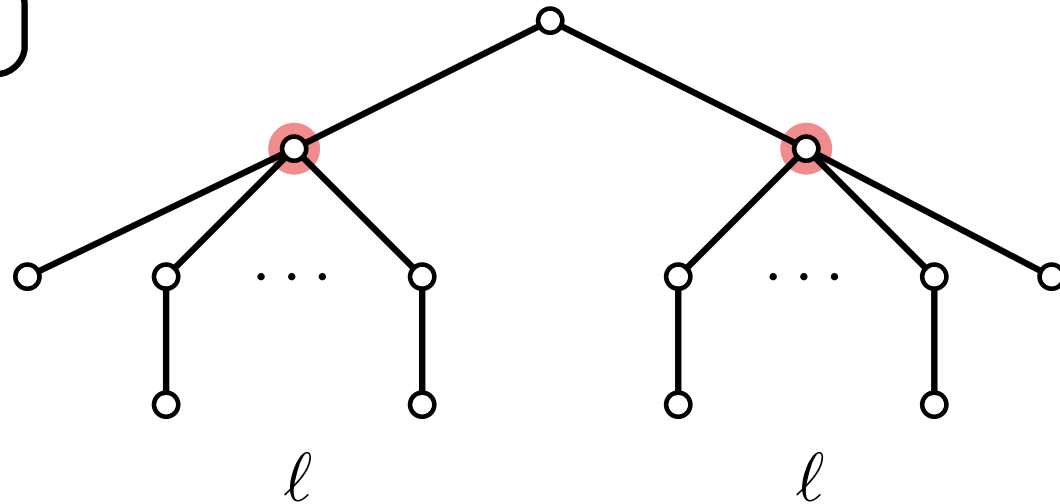
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

2



Greedy Heuristic

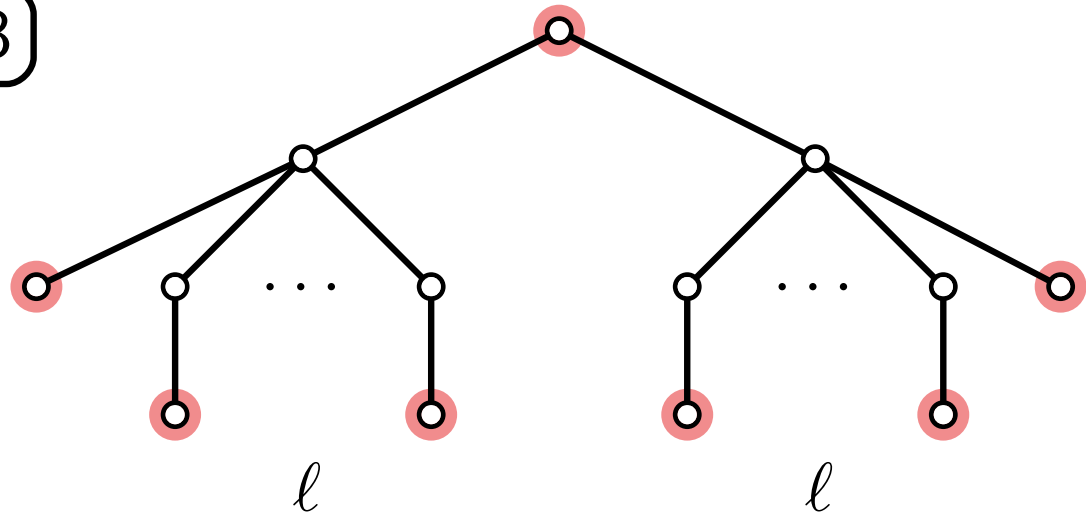
Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

$$2\ell + 3$$



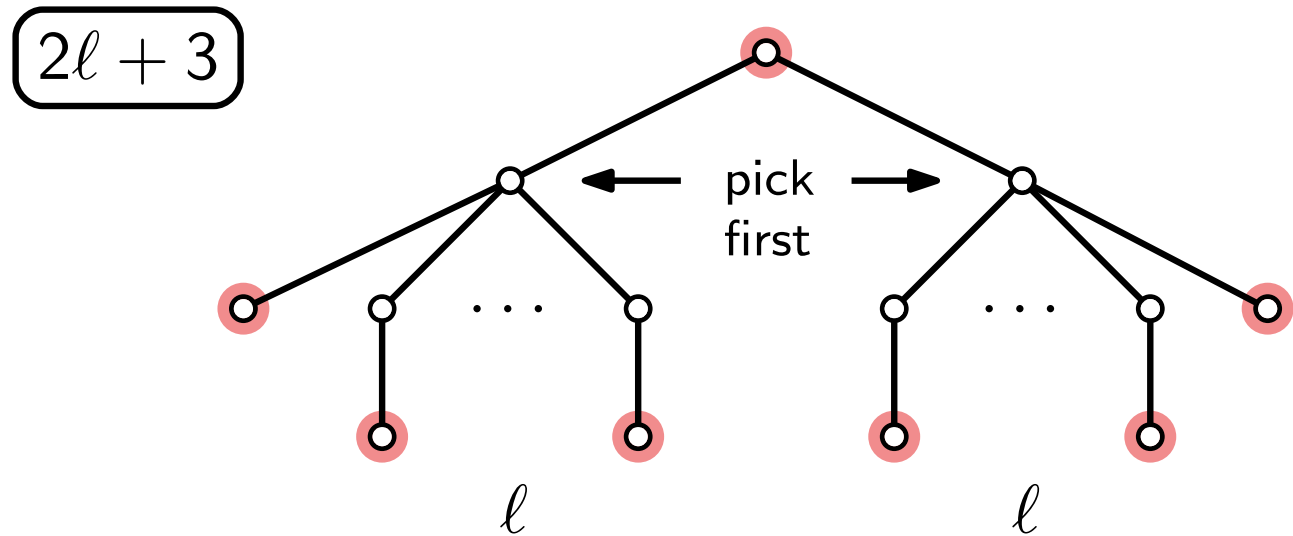
Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



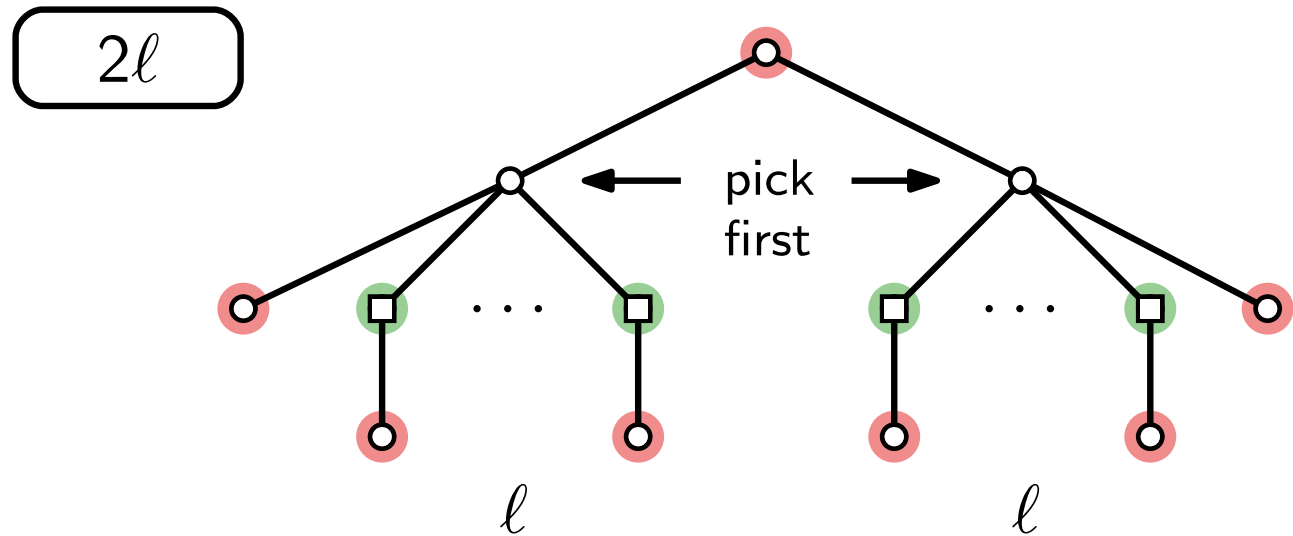
Greedy Heuristic

Simple Greedy

Phase 1: Select an initial frame.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



Greedy Heuristic

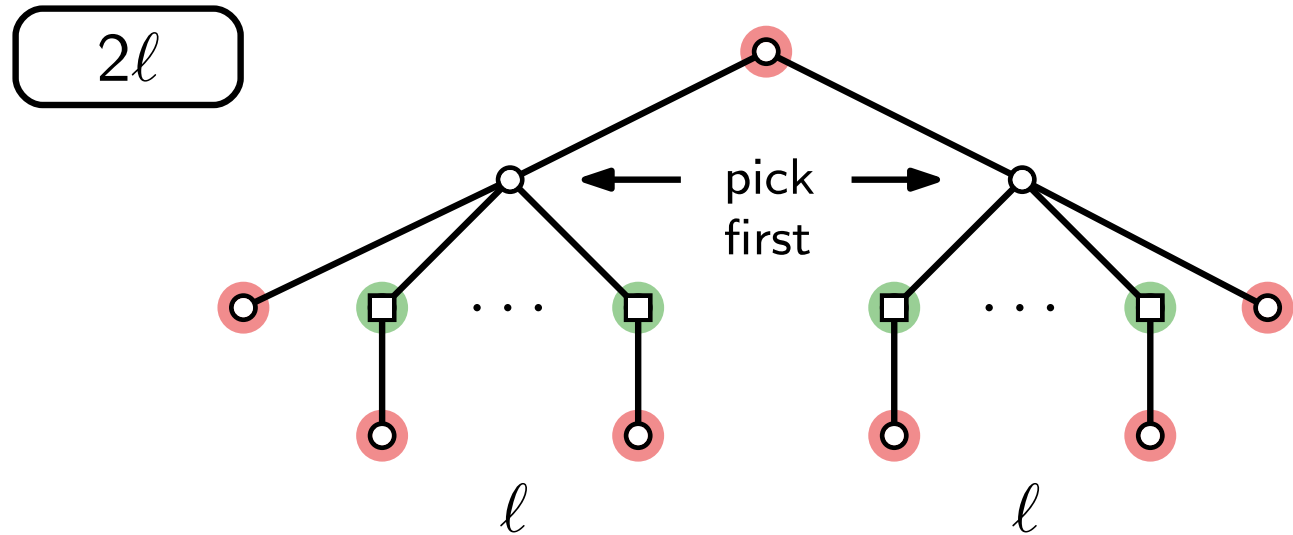
Advanced

~~Simple~~ Greedy

Phase 1: Select an initial frame **and final a frame**.

Phase 2: For each subsequent frame, pick a new edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



Greedy Heuristic

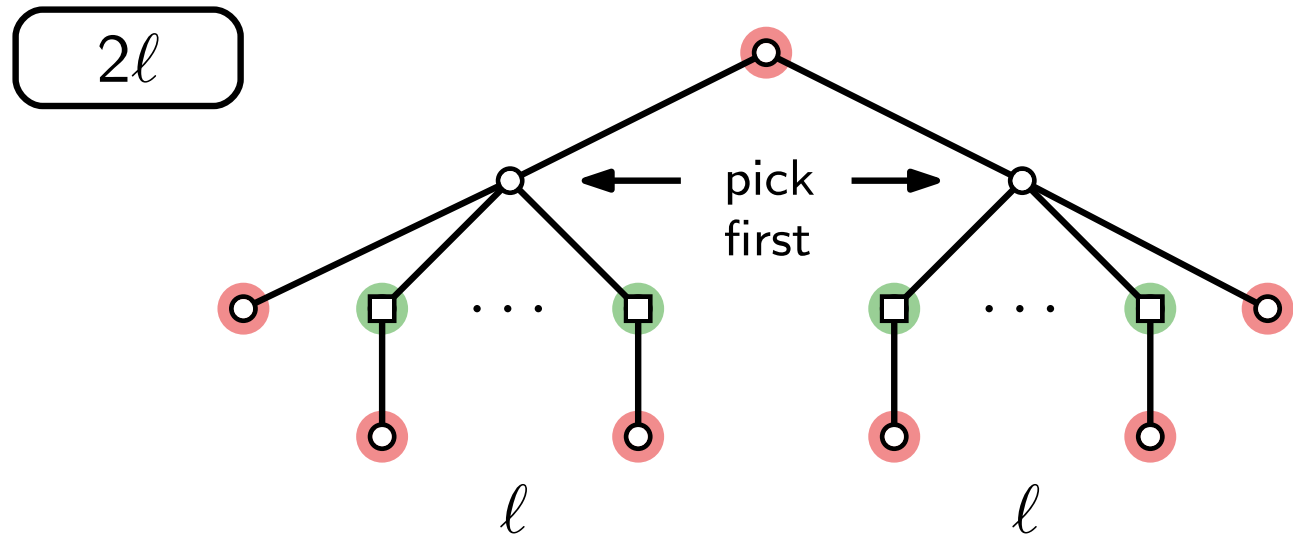
Advanced

~~Simple~~ Greedy

Phase 1: Select an initial frame and final a frame.

Phase 2: For each subsequent frame, pick a new
admissible edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



Greedy Heuristic

Advanced

~~Simple~~ Greedy

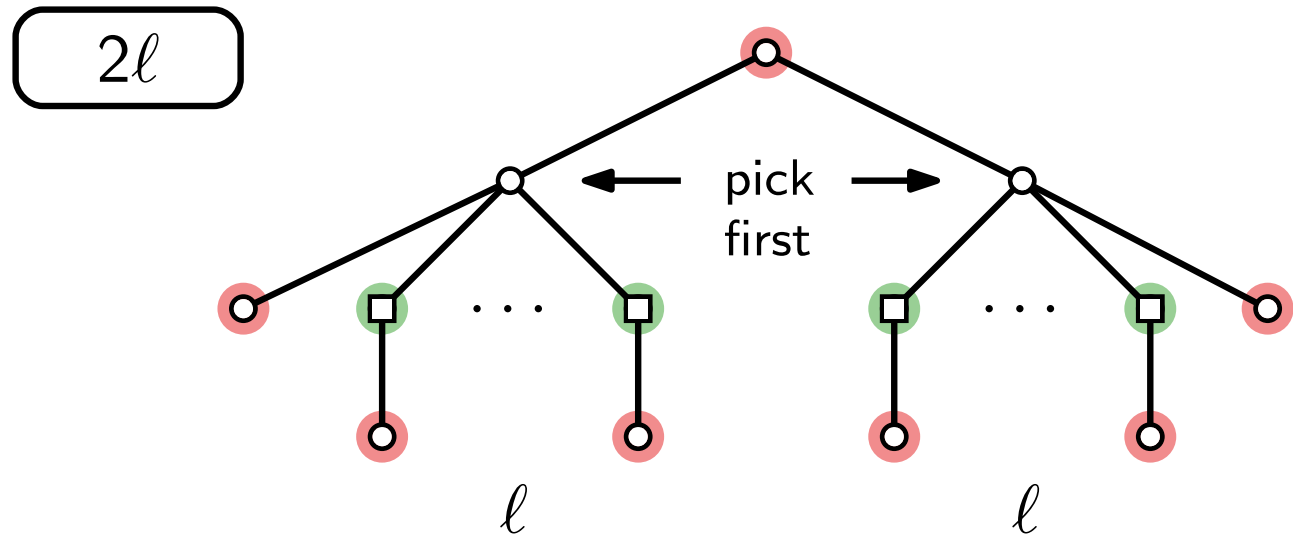
Phase 1: Select an initial frame **and final a frame**.

Phase 2: For each subsequent frame, pick a new **admissible** edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

A pair (I_1, I_2) of independent sets is *pareto-optimal* if $\min\{|I_1|, |I_2|\}$ and $\max\{|I_1|, |I_2|\}$ is maximum

Theorem. Given a graph G , it is NP-hard to compute a *pareto-optimal* pair of disjoint independent sets in G .



Greedy Heuristic

Advanced

~~Simple~~ Greedy

Phase 1: Select an initial frame **and final a frame**.

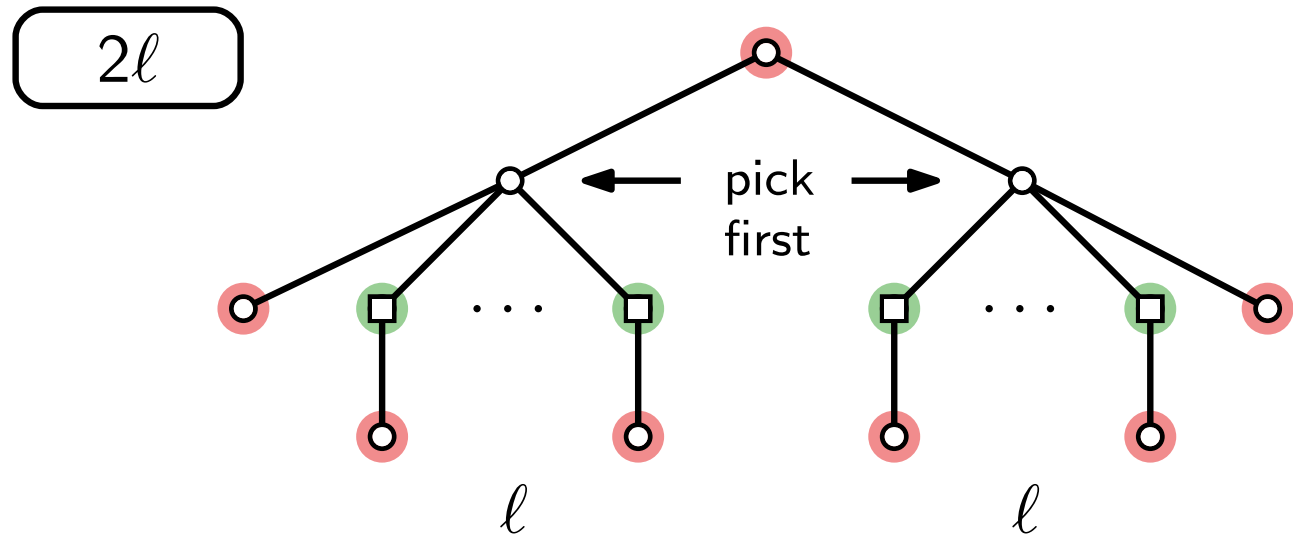
Phase 2: For each subsequent frame, pick a new **admissible** edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

A pair (I_1, I_2) of independent sets is *pareto-optimal* if $\min\{|I_1|, |I_2|\}$ and $\max\{|I_1|, |I_2|\}$ is maximum

Theorem. Given a graph G , it is NP-hard to compute a *pareto-optimal* pair of disjoint independent sets in G .

but FPT w.r.t. treewidth



Greedy Heuristic

Advanced

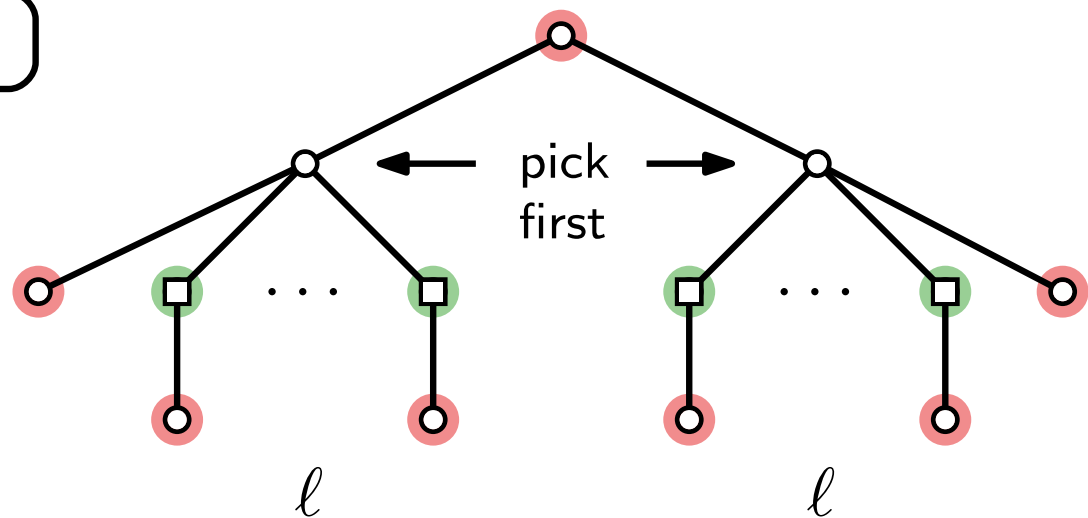
~~Simple~~ Greedy

Phase 1: Select an initial frame **and final a frame**.

Phase 2: For each subsequent frame, pick a new **admissible** edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.

2ℓ



A pair (I_1, I_2) of independent sets is *pareto-optimal* if $\min\{|I_1|, |I_2|\}$ and $\max\{|I_1|, |I_2|\}$ is maximum

Theorem. Given a graph G , it is NP-hard to compute a *pareto-optimal* pair of disjoint independent sets in G .

but FPT w.r.t. treewidth

Alternatives: $\nearrow$ w.r.t. crossing graph
Iteratively pick a vertex w with minimum degree for either I_1 or I_2 .

Greedy Heuristic

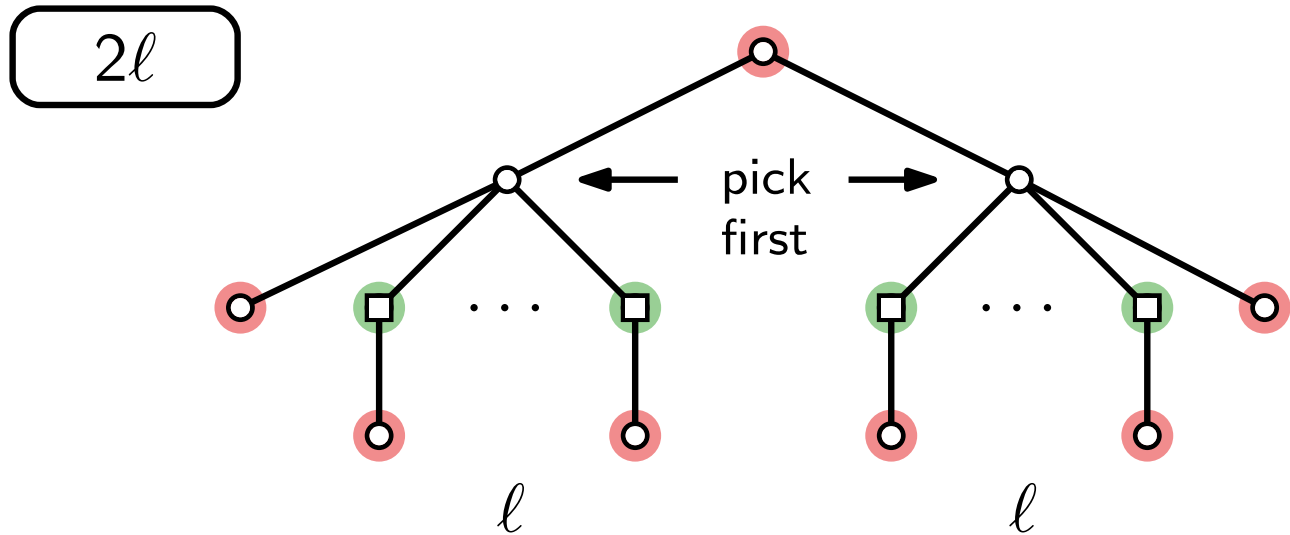
Advanced

~~Simple~~ Greedy

Phase 1: Select an initial frame **and final a frame.**

Phase 2: For each subsequent frame, pick a new **admissible** edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



A pair (I_1, I_2) of independent sets is *pareto-optimal* if $\min\{|I_1|, |I_2|\}$ and $\max\{|I_1|, |I_2|\}$ is maximum

Theorem. Given a graph G , it is NP-hard to compute a *pareto-optimal* pair of disjoint independent sets in G .

but FPT w.r.t. treewidth

Alternatives: $\nearrow$ w.r.t. crossing graph
Iteratively pick a vertex w with minimum degree for either I_1 or I_2 .

Two-Stage Approach $|I_1| \leq |E(G)|/2$
Create maximal independent set I_1 , then maximal independent set I_2 .

Greedy Heuristic

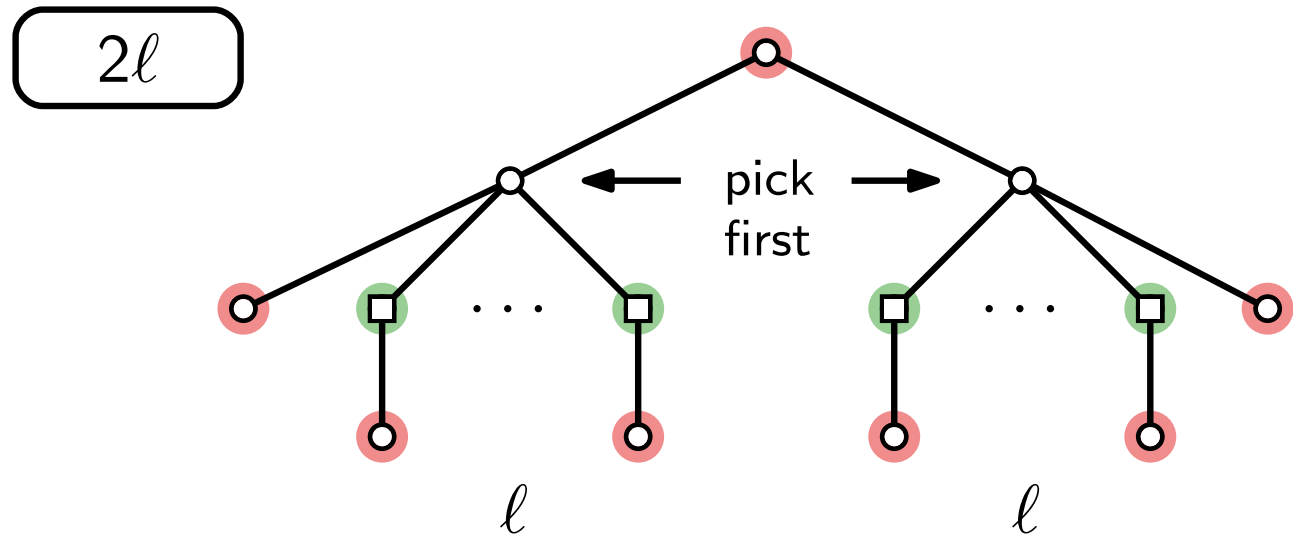
Advanced

~~Simple~~ Greedy

Phase 1: Select an initial frame **and final a frame**.

Phase 2: For each subsequent frame, pick a new **admissible** edge that causes the fewest edge deletions.

Observation: The initial frame will be grown to a maximal independent set.



A pair (I_1, I_2) of independent sets is *pareto-optimal* if $\min\{|I_1|, |I_2|\}$ and $\max\{|I_1|, |I_2|\}$ is maximum

Theorem. Given a graph G , it is NP-hard to compute a *pareto-optimal* pair of disjoint independent sets in G .

but FPT w.r.t. treewidth

Alternatives: $\rightarrow$ w.r.t. crossing graph
Iteratively pick a vertex w with minimum degree for either I_1 or I_2 .

Two-Stage Approach $|I_1| \leq |E(G)|/2$
Create maximal independent set I_1 , then maximal independent set I_2 .

Alternating Approach
Add w alternatingly between I_1 and I_2 .

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset: **Geometric Graphs:** **Crossing Graphs:**

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Geometric Graphs:

- Erdős–Rényi
 - Real graphs¹
- Fruchterman-Reingold

Crossing Graphs:

¹ <https://networkrepository.com>

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Geometric Graphs:

- Erdős–Rényi
 - Real graphs¹
- Fruchtermann-Reingold

Crossing Graphs:

- Trees
- Series-parallel
- Planar graphs

¹ <https://networkrepository.com>

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Geometric Graphs:

■ Erdős–Rényi

■ Real graphs¹

Fruchtermann-Reingold

Crossing Graphs:

■ Trees

■ Series-parallel

■ Planar graphs

synthetic

Graph sizes: synthetic: $n \in \{10, 20, \dots, 100\}$, real: $30 \leq n \leq 379$

Densities (synthetic): $d \in \{1.2, 1.4, 1.6, 2.0, 2.4\}$

Total instances: 637

¹ <https://networkrepository.com>

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Geometric Graphs:

■ Erdős–Rényi

■ Real graphs¹

Fruchtermann-Reingold

Crossing Graphs:

■ Trees

■ Series-parallel

■ Planar graphs

synthetic

Graph sizes: synthetic: $n \in \{10, 20, \dots, 100\}$, real: $30 \leq n \leq 379$

Densities (synthetic): $d \in \{1.2, 1.4, 1.6, 2.0, 2.4\}$

Total instances: 637

Exact: We used an ILP model to compare our heuristic with optimal solutions.

¹ <https://networkrepository.com>

Experiments: Setting

Goal: Evaluate the quality of the heuristic (and runtime).

Dataset:

Geometric Graphs:

■ Erdős–Rényi

■ Real graphs¹

Fruchtermann-Reingold

Crossing Graphs:

■ Trees

■ Series-parallel

■ Planar graphs

synthetic

Graph sizes: synthetic: $n \in \{10, 20, \dots, 100\}$, real: $30 \leq n \leq 379$

Densities (synthetic): $d \in \{1.2, 1.4, 1.6, 2.0, 2.4\}$

Total instances: 637

Exact: We used an ILP model to compare our heuristic with optimal solutions.

Time out: 15 mins (ILP)

¹ <https://networkrepository.com>

Experiments: Results – an Overview

- ILP solved 406/637 instances within the time limit (Gurobi)

Experiments: Results – an Overview

quality ratio: $\frac{ALG}{OPT}$

SD: standard deviation

- ILP solved 406/637 instances within the time limit (Gurobi)
- Quality of the Advanced Greedy variants (combined instances)

	% optimally solved	Quality ratio	Min–Max quality ratio	SD (σ)
Pareto-optimal	59.9%	94.14%	50.00% – 100.00%	9.87%
Two-Stage	5.7%	67.91%	33.33% – 100.00%	13.44%
Alternating	14.3%	82.76%	44.44% – 100.00%	10.36%

Experiments: Results – an Overview

quality ratio: $\frac{ALG}{OPT}$
SD: standard deviation

- ILP solved 406/637 instances within the time limit (Gurobi)
- Quality of the Advanced Greedy variants (combined instances)

	% optimally solved	Quality ratio	Min–Max quality ratio	SD (σ)
Pareto-optimal	59.9%	94.14%	50.00% – 100.00%	9.87%
Two-Stage	5.7%	67.91%	33.33% – 100.00%	13.44%
Alternating	14.3%	82.76%	44.44% – 100.00%	10.36%

Conclusion:

- Heuristic with pareto-optimal approach produces the best results but is impractical for inputs with crossing graphs with large treewidth.
- Heuristic with alternating approach produces good results and is efficient.

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is MAXMINFRAMEPLANARSTORY NP-hard for 3-planar graphs?

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is `MAXMINFRAMEPLANARSTORY` NP-hard for 3-planar graphs?
- Parameterized Complexity?

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is MAXMINFRAMEPLANARSTORY NP-hard for 3-planar graphs?
- Parameterized Complexity?

Practice

- Improve ILP formulation.

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is `MAXMINFRAMEPLANARSTORY` NP-hard for 3-planar graphs?
- Parameterized Complexity?

Practice

- Improve ILP formulation.
- Find better heuristics for finding pairs of independent sets.

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is `MAXMINFRAMEPLANARSTORY` NP-hard for 3-planar graphs?
- Parameterized Complexity?

Practice

- Improve ILP formulation.
- Find better heuristics for finding pairs of independent sets.
- Change the model: Allow multiple edges to enter a frame.

Future Work

Theory

- Is Advanced Greedy with a pareto-optimal pair a constant-factor approximation?
- Is `MAXMINFRAMEPLANARSTORY` NP-hard for 3-planar graphs?
- Parameterized Complexity?

Practice

- Improve ILP formulation.
- Find better heuristics for finding pairs of independent sets.
- Change the model: Allow multiple edges to enter a frame.

Thank you :)